6 Normal modes of oscillation

6.1 A string fixed at both ends

We saw in the previous chapter that when a wave propagates along a string with one end fixed, the superposition of the incident wave and the reflected wave is a standing wave, which appears not to propagate at all. What would happen if *both* ends of the string were fixed?

This is not just a casual "what-if" question. All stringed musical instruments are based on this effect, and indeed as we shall see, wind instruments as well. And that's just waves in 1 dimension. What vibrations can arise in a 3-dimensional object is an important question in engineering.

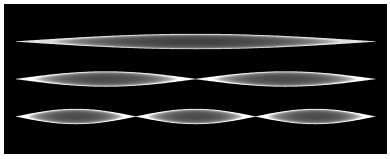


Figure 6.1: Some possible modes of oscillation of a string fixed at both ends.

Figure 6.1 shows some possible results. A standing wave is formed, such that the length of the string exactly matches an integer multiple of half a wavelength.

To see why this is so, recall that with one end fixed at $x = 0$, the string could sustain a standing wave of the form

$$y = A \sin \omega t \sin kx \tag{6.1}$$

where the amplitude A is twice the amplitude of the incident wave.

By fixing the other end of the string, at $x = L$, we are adding a constraint to the solution, namely that

$$\sin kL = 0 \tag{6.2}$$

which makes $y = 0$ at all times. This is possible if $kL = n\pi$, where $n = 1, 2, 3, \ldots$ Now, since $k = 2\pi/\lambda$, this means that

$$L = n\frac{\lambda}{2} \tag{6.3}$$

So the possible wavelengths that the string can sustain are

$$\lambda_n = \frac{2L}{n} \tag{6.4}$$

with $n = 1, 2, 3, \ldots$

Furthermore, if the possible wavelengths are constrained to a discrete set of values, then so are the frequencies, because the propagation speed

is determined only by the tension in the string and the linear mass density, and $v = f\lambda$. So the possible frequencies are

$$f_n = n\frac{v}{2L} \tag{6.5}$$

It is possible to test this experimentally by forcing a string to vibrate at a certain frequency. If the frequency does not match one of the values given by equation (6.5), then the wave quickly dies down and the string hardly vibrates at all. But if it does, then the string vibrates quite visibly. So our theory really describes waves in a fixed string that can be sustained over a long time. We generally refer to this as a **steady state**.

Each possible wave is called a **mode** of vibration of the string. The mode with the lowest frequency is called the **fundamental mode**. The modes of a vibrating string are also known as **harmonics**. The fundamental mode is also the first harmonic.

In a musical instrument like a guitar, the string is plucked, which actually sets waves with a wide range of frequencies propagating along the string and reflecting off the fixed ends. In the steady state, only the frequencies with values given by (6.5) survive.

Figure 6.2 shows the **spectrum** obtained from a single note played on an acoustic guitar. To produce a spectrum, a sound is broken down into its component frequencies, and the intensity of the sound at each frequency is plotted as a function of frequency.[1] Equally spaced peaks are clearly visible, corresponding to the different harmonics. It's interesting to note also that the intensities of the different harmonics are not the same; broadly speaking, the intensities tend to diminish with increasing frequency. In fact, the relative intensities of the harmonics are characteristic of the instrument. The brain uses this information to determine which instrument produced a certain sound.

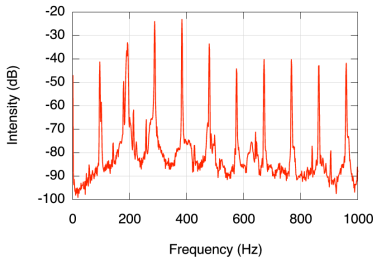


Figure 6.2: Guitar spectrum

[1] Notice that the intensity is measured in dB. The intensity is actually calculated as a fraction of the full range in the particular recording, and the ratio is measured in dB. So the actual numerical values are not very meaningful in themselves, but can be used to compare the intensities at different frequencies, which is the purpose of a spectrum plot.

6.2 *A tube with an open end*

As anybody who has tried blowing into a bottle can testify, you can also produce musical sounds in a cavity filled with air. The mechanism is the same as in a stretched string. Standing waves of certain frequencies are created in the cavity.

Consider a narrow tube with one end open. A sound wave travelling towards the closed end will of course be entirely reflected. But the surprising thing is that a wave travelling towards the open end *will also be (nearly) totally reflected*. This is because there is an abrupt change in the acoustic impedance at the open end.

In a tube, the velocity of the air immediately in contact with the inner wall is zero. The velocity gradually increases towards the centre of the tube. If the tube is narrow, the air cannot move very fast, even at

the centre, and therefore the acoustic impedance is high. Outside the tube, there are no such constraints against motion of the air, and the acoustic impedance is essentially zero. If medium 1 is the inside of the tube, where a wave is travelling towards the open end, and medium 2 is the outside air, we have the case where $Z_2 \ll Z_1$. As we saw in the previous chapter, almost all the energy in the incident wave is reflected back into the tube.

So, as with a string, a wave inside the tube bounces back and forth between the ends, even when one of the ends is open. The big difference between the open end and the closed one is that at the open end, the displacement of the air is maximum, whereas the displacement of the air at the closed end is exactly zero. We found that for an open end at $x = 0$, the form of the total wave in medium 1 was

$$y = A \cos \omega t \cos kx \tag{6.6}$$

where the amplitude A is again twice that of the incident wave alone. Now the fact that the other end is closed imposes an additional condition. If the length of the tube is L, and the displacement y is zero there at all times, we must have

$$\cos kL = 0 \Rightarrow kL = (2n-1)\frac{\pi}{2} \tag{6.7}$$

where $n = 1, 2, \ldots$ The wavelengths that survive in the steady state are given by

$$\lambda_n = \frac{4L}{2n-1} \tag{6.8}$$

with $n = 1, 2, \ldots$ The corresponding frequencies are

$$f_n = \frac{v}{\lambda_n} = (2n-1)\frac{v}{4L} \tag{6.9}$$

It's easy to visualize a standing wave in a string by taking a picture of the string with a long exposure time, as illustrated by the pictures in Figure 6.1. But how can we visualize the displacement of the air in a tube? What we can do is to plot a series of snapshots of the displacement versus position along the tube. If we do this for a narrow tube with one end open, we get the images shown in Figures 6.3–6.5.

Figure 6.6 shows the spectrum of sound produced by a test tube, which is narrow and open only at one end. Notice that this time the peaks are equally spaced, *with the exception of the fundamental mode*, which has a frequency that is only 1/2 the frequency of the second harmonic. This is quite consistent with equation (6.9).

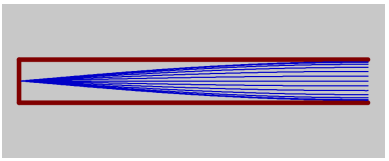


Figure 6.3: Tube with one end open – fundamental mode. Bear in mind that these are *longitudinal* waves: The vertical direction represents the longitudinal displacement of the air layers.

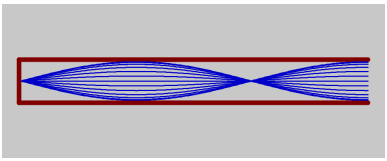


Figure 6.4: Tube with one end open – 2nd. harmonic.

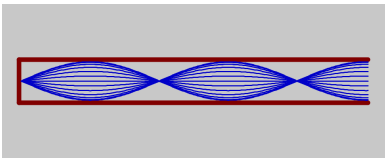


Figure 6.5: Tube with one end open – 3rd. harmonic.

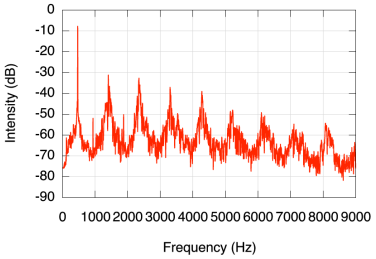


Figure 6.6: Spectrum of sound produced by a tube open at one end.

Example 6.1: *If the speed of sound is* $340\,\mathrm{m\,s^{-1}}$*, what is the approximate length of the tube of Figure 6.9?*

The difference between two successive harmonics is about 1000 Hz, and according to equation (6.9) this is equal to $v/(2L)$, from which we estimate $L \approx 17\,\mathrm{cm}$.

6.3 *A tube with both ends open*

The rule is simple: The displacement is zero at a closed end, and maximum at an open end. In other words, there is a node at a closed end and an antinode at the open end. So if both ends are open, there must be an antinode at each end. Figures 6.7–6.9 show the first three harmonics.

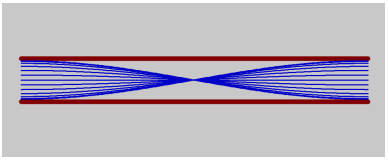


Figure 6.7: Tube with both ends open – Fundamental mode.

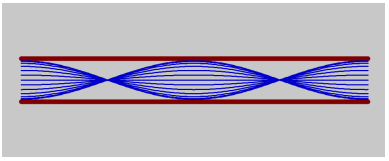


Figure 6.8: Tube with both ends open – Second harmonic.

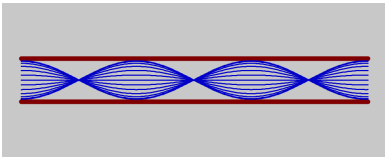


Figure 6.9: Tube with both ends open – Third harmonic

From the figures, we can infer the relation between the order of the harmonic n, the length of the tube L and the wavelength λ:

n	L/λ
1	1/2
2	1
3	3/2

The general rule is that

$$L = n\frac{\lambda_n}{2} \Rightarrow \lambda_n = \frac{2L}{n} \tag{6.10}$$

and therefore

$$f_n = n\frac{v}{2L} \tag{6.11}$$

This is the same spectrum as for a string fixed at both ends (or, suspending disbelief, a cavity closed at both ends).

Many wind instruments are effectively tubes open at both ends. To give an example, Figure 6.10 shows a portion of the spectrum produced by a French horn.

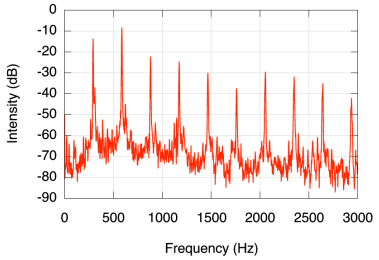


Figure 6.10: Spectrum of sound from a French horn.

Example 6.2: *What is the approximate length of the cavity of the French horn of Figure 6.10?*

This time the frequency difference between successive harmonics is about 300 Hz, which should be equal to $v/(2L)$ according to equation (6.11). With $v = 340\,\mathrm{m\,s^{-1}}$, this gives $L \approx 57\,\mathrm{cm}$. Interestingly, a French horn is a lot more compact, because the cavity is coiled in a circle, as Figure 6.11 shows.



Figure 6.11: A French horn (BenP).